User Onboarding Funnel Analysis

Understand. Optimize. Convert. — A Case Study by Anthonia Ozobialu

Project Overview

Most products lose the majority of their users somewhere between first visit and final conversion, but where exactly and why is rarely obvious without breaking the journey into stages. This case study analyzes a 90,400-user e-commerce onboarding funnel to quantify conversion and drop-off at each step, segment behavior by device and demographic, and identify the single biggest opportunity for improvement.

Business question: Where in the user journey are we losing the most people, and what should we fix first?

Objectives

- Load and clean raw funnel event data across five source tables
- Calculate conversion % and drop-off % at every stage of the funnel
- Segment funnel performance by device and sex to spot behavioral differences
- Identify the single largest bottleneck in the journey
- Build an interactive Power BI dashboard with slicers and drill-downs
- Translate findings into concrete, prioritized recommendations

Dataset

Source: Sales Funnel Data - User Drop-off Analysis (Kaggle, by andrewjayasatyo)

Table	Description
home_page_table.csv	Users who landed on the homepage
search_page_table.csv	Users who reached the search page
payment_page_table.csv	Users who reached the payment page
payment_confirmation_table.csv	Users who completed a purchase
user_table.csv	User-level demographics: device and sex

Scale: 90,400 total users tracked across the full journey.

Methodology

1. **Data loading & merging** — Combined the five stage tables with the user demographics table on user_id to build a single funnel-stage dataset per user.

2. Funnel metric calculation — For each stage, computed users reaching that stage, conversion % (users at stage / users at Home), and drop-off % (1 - users at stage / users at previous stage).

3. Segmentation — Broke down conversion and drop-off by Device (Desktop/Mobile) and Sex to check whether the bottleneck was uniform or concentrated in a specific group.

4. Bottleneck identification — Compared drop-off percentages across all stage transitions to find the point of maximum loss.

5. Dashboard build — Designed a Power BI report with KPI cards, funnel/bar visuals, a conversion gauge against an 80% target, a detail table, and interactive Date/Device/Sex slicers.

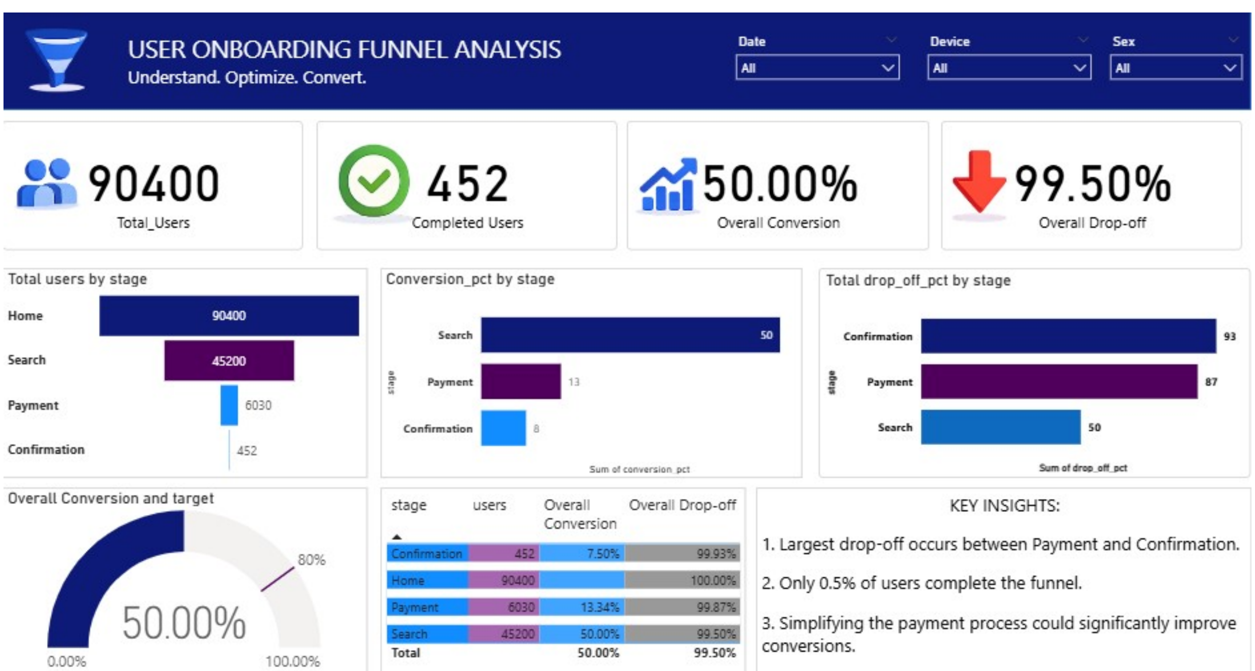
Key Findings

Stage	Users	Overall Conversion	Overall Drop-off
Home	90,400	100.00%	-
Search	45,200	50.00%	50.00%
Payment	6,030	13.34%	99.87%
Confirmation	452	7.50%	99.93%

Note: "Overall" figures are measured relative to total Home users; stage-to-stage drop-off (below) shows the true size of each jump.

1. **Half of all users never make it past Search** — the Home → Search transition alone accounts for a 50% drop-off.
2. **The single largest drop happens between Payment and Confirmation** (93% drop-off at that step) — nearly everyone who reaches checkout still fails to complete it.
3. **Only 0.5% of the original 90,400 users complete the entire funnel** (452 users), against an 80% conversion target.
4. Overall conversion sits at **50.00%**, but that figure is driven almost entirely by the first step; downstream stages are where the real leakage happens.

Dashboard



Power BI dashboard: User Onboarding Funnel Analysis, with KPI cards, stage-by-stage funnel, conversion/drop-off breakdowns, a conversion gauge, and Date/Device/Sex slicers.

Recommendations

1. Prioritize fixing the Payment → Confirmation step. With a 93% drop-off, this is the highest-leverage stage. Investigate friction points: payment method limitations, form complexity, unexpected fees, or trust/security signals at checkout.

2. Simplify the payment process. Reduce the number of fields/steps required, add guest checkout, and support more payment methods to reduce abandonment at this stage.

3. Investigate the Home → Search drop-off. Losing half of all visitors before they even search suggests onboarding friction or unclear value proposition on the landing page — worth A/B testing homepage CTAs and layout.

4. Use segment-specific interventions. Use the device/sex breakdown to check whether one segment (e.g. mobile users) disproportionately drops off, and tailor UX fixes accordingly.

5. Set stage-specific targets, not just an overall conversion target. Tracking Search-rate and Payment-completion-rate separately will make it easier to detect which fix is working.

Tools & Techniques

- **Python (pandas)** — data loading, merging, funnel metric calculation
- **Power BI** — interactive dashboard with KPI cards, funnel/bar charts, gauge visual, and Date/Device/Sex slicers
- **SQL** — stage-wise aggregation and conversion logic

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